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Lab 1.2 IoT Sensor Data Collection 1 - Temperature Sensor

1.1 Plotting the sensor data from the DS18B20

Plot the measured data from the CSV file using the template code below and attach the two plots (temperature according to timestamp).

Lets plot the temperature graph first.

Import essential libraries:

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import io
import seaborn as sns
```

Load CSV Data

The following code allows you to upload the data stored in a csv file to this Colab notebook.

```
#This code allows you to upload a file from your local drive
from google.colab import files
uploaded = files.upload()
#If you will work with excel, please save your file with a .CSV format
#Check the output file name below. You will use that file name in the
```

Choose Files ds18b20_data.csv

ds18b20_data.csv(text/csv) - 1365 bytes, last modified: 2/4/2026 - 100% done

```
Saving ds18b20_data.csv to ds18b20_data (2).csv
```

```
#Write the file name from the output above, inside the ' ' within the bracke
#It will sub where it reads 'yourfilename.csv'
#If you want to check the uploaded file, click Files on the left-hand side m

df
```

	Timestamp	Temperature(°C)	Temperature(°F)	time
0	2026-02-04 13:25:51	23.69	74.64	2026-02-04 13:25:51
1	2026-02-04 13:25:52	23.75	74.75	2026-02-04 13:25:52
2	2026-02-04 13:25:54	23.75	74.75	2026-02-04 13:25:54
3	2026-02-04 13:25:56	23.75	74.75	2026-02-04 13:25:56
4	2026-02-04 13:25:58	23.75	74.75	2026-02-04 13:25:58
5	2026-02-04 13:26:00	23.75	74.75	2026-02-04 13:26:00
6	2026-02-04 13:26:01	23.88	74.97	2026-02-04 13:26:01
7	2026-02-04 13:26:03	24.44	75.99	2026-02-04 13:26:03
8	2026-02-04 13:26:05	25.19	77.34	2026-02-04 13:26:05
9	2026-02-04 13:26:07	25.88	78.58	2026-02-04 13:26:07
10	2026-02-04 13:26:09	26.50	79.70	2026-02-04 13:26:09
11	2026-02-04 13:26:10	27.31	81.16	2026-02-04 13:26:10
12	2026-02-04 13:26:12	28.00	82.40	2026-02-04 13:26:12
13	2026-02-04 13:26:14	28.56	83.41	2026-02-04 13:26:14
14	2026-02-04 13:26:16	29.00	84.20	2026-02-04 13:26:16
15	2026-02-04 13:26:18	29.44	84.99	2026-02-04 13:26:18
16	2026-02-04 13:26:19	29.75	85.55	2026-02-04 13:26:19
17	2026-02-04 13:26:21	30.00	86.00	2026-02-04 13:26:21
18	2026-02-04 13:26:23	30.25	86.45	2026-02-04 13:26:23
19	2026-02-04 13:26:25	30.50	86.90	2026-02-04 13:26:25
20	2026-02-04 13:26:27	30.69	87.24	2026-02-04 13:26:27

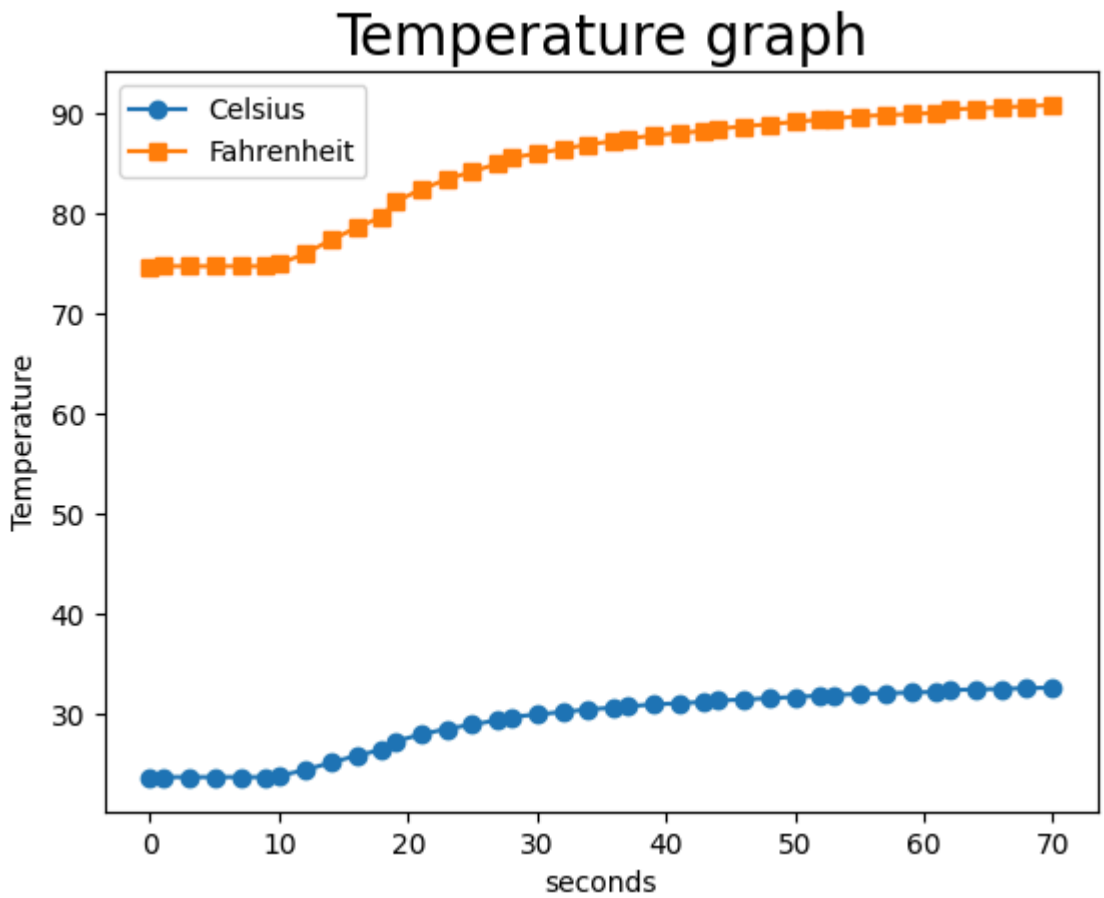
Next steps: Generate code with df New interactive sheet

Data plotting

Plot a chart that includes temperature (in °C, °F) in the code box below:

```
df['time'] = pd.to_datetime(df["Timestamp"])
time = df['time'].dt.hour*3600 + df['time'].dt.minute*60 + df['time'].dt.sec
t1 = time - time[0]
temperature1 = df['Temperature(°C)']
temperature2 = df['Temperature(°F)']


fig = plt.figure()

ax = fig.add_subplot(111)

ax.set_title('Temperature graph',fontsize= 20) # title of plot

ax.set_xlabel('seconds',fontsize = 10) #xlabel
ax.set_ylabel('Temperature', fontsize=10)

ax.plot(t1, temperature1, '-o', label='Celsius')
ax.plot(t1, temperature2, '-s', label='Fahrenheit')

ax.legend()

plt.show()
                    # prints the graph
```

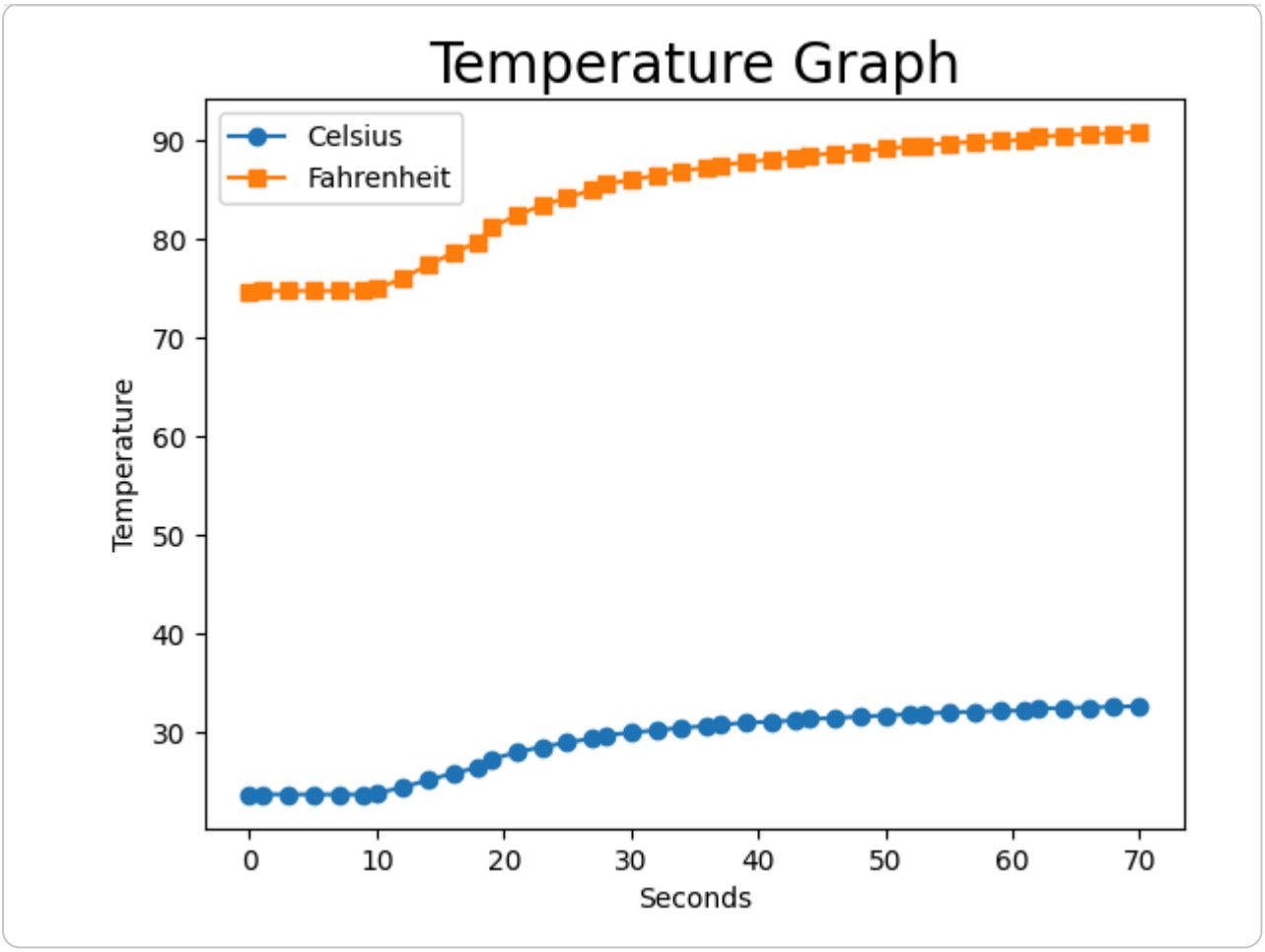
21	2026-02-04 13:26:28	30.81	87.46	2026-02-04 13:26:28
22	2026-02-04 13:26:30	31.00	87.80	2026-02-04 13:26:30
32	13:26:48	32.12	89.83	13:26:48
33	2026-02-04 13:26:50	32.19	89.94	2026-02-04 13:26:50
34	2026-02-04 13:26:52	32.25	90.05	2026-02-04 13:26:52
35	2026-02-04 13:26:53	32.44	90.39	2026-02-04 13:26:53
36	2026-02-04 13:26:55	32.50	90.50	2026-02-04 13:26:55
37	2026-02-04 13:26:57	32.56	90.61	2026-02-04 13:26:57
38	2026-02-04 13:26:59	32.62	90.72	2026-02-04 13:26:59
39	2026-02-04 13:27:01	32.69	90.84	2026-02-04 13:27:01



Task 2.1

Plot a chart using that includes both temperature unit in the code box below:

```
df['time'] = pd.to_datetime(df["Timestamp"])
time = df['time'].dt.hour*3600 + df['time'].dt.minute*60 + df['time'].dt.sec
t1 = time - time.iloc[0]

temperature1 = df['Temperature(°C)']
temperature2 = df['Temperature(°F)']

fig = plt.figure()
ax = fig.add_subplot(111)

ax.set_title('Temperature Graph', fontsize=20)
ax.set_xlabel('Seconds', fontsize=10)
ax.set_ylabel('Temperature', fontsize=10)

ax.plot(t1, temperature1, '-o', label='Celsius')
ax.plot(t1, temperature2, '-s', label='Fahrenheit')

ax.legend()

plt.show()
```



Lab1 Summary and Deliverables

Answer the following questions for your achievements

Q1. Please summarize Lab1

We learned how to interface the DS18B20 temperature sensor with a Raspberry Pi using correct GPIO pin connections and 1-Wire communication. We wrote Python code to read temperature values in both Celsius and Fahrenheit along with timestamps. The collected data was saved into a CSV file, and that file was then used to generate a Time vs Temperature graph for analysis.

Q2. What skills did you have to develop to accomplish this project?

Embedded systems interfacing

Sensor integration

Python programming

Data logging & CSV handling

Data visualization and analysis

Q3. What aspects of this project were the most beneficial for your learning?

The project strengthened my understanding of embedded systems, practical programming, and data analysis in a very applied way.

Q4. What challenges did you encounter in completing the project?

The project required patience in debugging both hardware and software together.

Q5. How did you overcome the challenges or remedy the problems encountered?

I overcame the challenges with the help of my instructor and by using online resources.